

Algorithmic Pseudocode Sample 21

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

Require:

τ : Scale of coordinate noise

GNN_θ : Graph Neural Network with parameter θ

NoiseHead $_{\theta_n}$: Network module with parameter θ_n for prediction of node-level noise of each atom

LabelHead $_{\theta_l}$: Network module with parameter θ_l for prediction of graph-level label of x_i

X : Training dataset

x_i : Input conformation

y_i : Label of x_i

T : Training steps

$\mathcal{N}$: Gaussian distribution

λ_p : Loss weight of property prediction loss

λ_n : Loss weight of Noisy Nodes loss

```
1: while  $T \neq 0$  do
2:    $x_i, y_i = \text{dataloader}(X)$   $\triangleright$  random sample  $x_i$  and corresponding label  $y_i$  from  $X$ 
3:    $\tilde{x} = x_i + \Delta x_i$ , where  $\Delta x_i \sim \mathcal{N}(0, \tau^2 I_{3N})$ ,  $N$  is atom number of  $x_i$ 
4:    $y_i^{pred} = \text{LabelHead}_{\theta_l}(GNN_\theta(\tilde{x}))$ 
5:    $\Delta x_i^{pred} = \text{NoiseHead}_{\theta_n}(GNN_\theta(\tilde{x}))$ 
6:    $\text{Loss} = \lambda_p \text{PropertyPredictionLoss}(y_i^{pred}, y_i) + \lambda_n ||\Delta x_i^{pred} - \Delta x_i||_2^2$ 
7:    $\text{Optimise}(\text{Loss})$ 
8:    $T = T - 1$ 
9: end while
```
